Nouh Taha CHEBCHOUB

Applied AI / ML Engineer – Data Scientist
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Professional Summary

Applied AI / ML Engineer and Data Scientist with a master's in Data Science and around two years of industry and research experience across computer vision, NLP, LLMs, and deep learning. Experienced in building end-to-end ML pipelines, evaluating and refining models, and taking AI applications from experimentation to production deployment.

Professional Experience

Data & AI Engineer, Variable Data Co. Jul 2025 – Aug 2026
Saudi Arabia (Remote)

- Built and deployed a Document AI system for scanned and native medical, motor, and general insurance PDFs, combining OCR, LLM-based extraction, and rule-based validation.
- Owned model selection, prompt design, API serving, and deployment; refined extraction and report quality through iterative client testing.
- Reduced report turnaround from over two days of manual work to under 15 minutes, with high report quality reported by the client during testing.

Data Scientist Intern (Automotive / EV), DHBW Feb 2025 – Jul 2025
Friedrichshafen, Germany

- Designed and deployed an electric vehicle digital twin using CARLA, integrating real-time telemetry for predictive analytics and maintenance.
- Developed a physics-informed neural network for battery state-of-charge estimation, embedding the Coulomb counting differential equation into the loss function to enforce physical consistency.

Machine Learning Engineer Intern, Cadi Ayyad University Nov 2024 – Mar 2025
Faculty of Sciences Semlalia (FSSM), Marrakech, Morocco

- Benchmarked CNN, GRU, and Bi-LSTM models for EEG-based emotion recognition on DEAP; completed research and manuscript preparation through publication in *Sensors* in March 2025.

Projects

TwinFlux: End-to-End EV Energy Data & ML Platform

- Built multi-source vehicle data pipelines, PostgreSQL and object storage, a physics-calibrated simulator, and MLflow-tracked models; validated state-of-charge estimates against real-world trips.
- *Stack:* Python, SQL, FastAPI, PostgreSQL, Redis, MinIO, Docker, GitHub Actions, MLflow, Grafana.

GAN-Sim: GAN-Based Aim Assistance in CS:GO

- Combined YOLO detection with human-like aiming movements; curated and annotated images and built the detection and generative modelling pipeline.
- *Stack:* Python, TensorFlow, OpenCV, YOLO.

Skills

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, CNNs, RNNs, GANs, physics-informed neural networks, transformers, model evaluation.
NLP, Generative AI & Vision: LLMs, RAG, LangChain, LangGraph, prompt engineering, OCR, Document AI, OpenCV, YOLO.
Engineering & MLOps: Python, SQL, PySpark, FastAPI, Docker, Git, GitHub Actions, MLflow, PostgreSQL, MongoDB, Redis, Oracle Cloud, Grafana, Prometheus.
Languages: French (Fluent), English (Fluent), Arabic (Native).

Education

Master's Degree in Data Science 2023 – 2025
Cadi Ayyad University, FSSM – Marrakech, Morocco

Bachelor's Degree in Mathematics and IT 2020 – 2023
Cadi Ayyad University, FSSM – Marrakech, Morocco

Publications

Enhancing EEG-based Emotion Detection with Hybrid Models, *Sensors* (MDPI), 2025. DOI: 10.3390/s25061827

Certifications

Machine Learning – DeepLearning.AI · **Python for Data Science, AI & Development** – IBM